

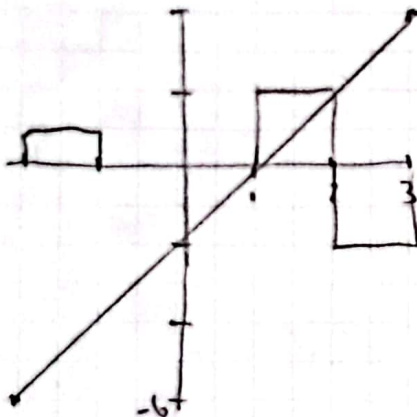
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1 $x(t) = u(t+2) - u(t+2)$

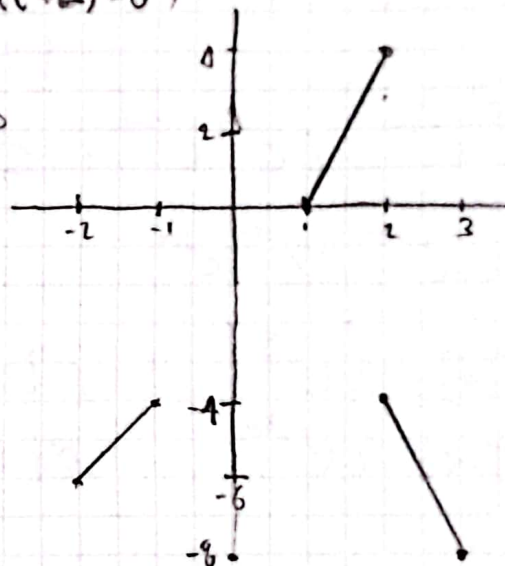
$y(t) = u(t-3) - u(t-2) + u(t-2) - u(t-1)$

2. Graficar $w(t) = z(t) + r(2(t+2) - 6)$



Multiplicando

$\Rightarrow$



$w(t)$

3. Encontrar la transformada de Fourier $K = 2$

$x(t) = 4 \cos(8\pi t + (\frac{\pi}{4})) + 2 \sin(4\pi t) + 5 =$

i) $\omega_1 = 8\pi = \frac{2\pi}{T_1} \quad T_1 = \frac{1}{4}$

$\omega_2 = 4\pi = \frac{2\pi}{T_2} \quad T_2 = \frac{1}{2}$

$\frac{T_1}{T_2} = \frac{1}{2}$ (es periodico)

Periodo:

$\frac{1}{2} \mid \frac{1}{4} \mid \frac{2}{4} \mid \frac{3}{4} \mid \frac{4}{4}$
 $\frac{1}{4} \mid \frac{1}{4} \mid \frac{2}{4} \mid \frac{3}{4} \mid \frac{4}{4}$

mcm = 1

$$4 \cos(8\pi t + \pi) + 2 \sin(4\pi t) + 5 = \sum_{k=-\infty}^{\infty} X[k] e^{jk2\pi t}$$

$$2 \left(\frac{e^{j(8\pi t + \pi)} + e^{-j(8\pi t + \pi)}}{2} \right) + \frac{e^{j4\pi t} - e^{-j4\pi t}}{j} + 5$$

$$2 e^{j8\pi t} e^{j\pi} + 2 e^{-j8\pi t} e^{-j\pi} + \frac{e^{j4\pi t}}{j} - \frac{e^{-j4\pi t}}{j} + 5 = \sum_{k=-\infty}^{\infty} X[k]$$

$$X[4] e^{j8\pi t} = 2 e^{j8\pi t} e^{j\pi}$$

$$X[4] = 2 e^{j\pi}$$

$$X[-4] = 2 e^{-j\pi}$$

$$X[2] = \frac{1}{j}$$

$$X[-2] = -\frac{1}{j}$$

$$X[0] = 5$$

$$X[k] = \left\{ 2e^{-j\pi}, 0, -\frac{1}{j}, 0, 5, 0, \frac{1}{j}, 0, 2e^{-j\pi} \right\}$$

7. Señal definida entre a Hz y b Hz $f_s = 100$ Hz

- Diseñar pre-amplificador si se requiere que la banda de rechazo sea 10 veces mayor amplitud a la señal de entrada

$$a = 5 \quad b = 50$$

$$20 \log(10) = 20 \text{ dB}$$